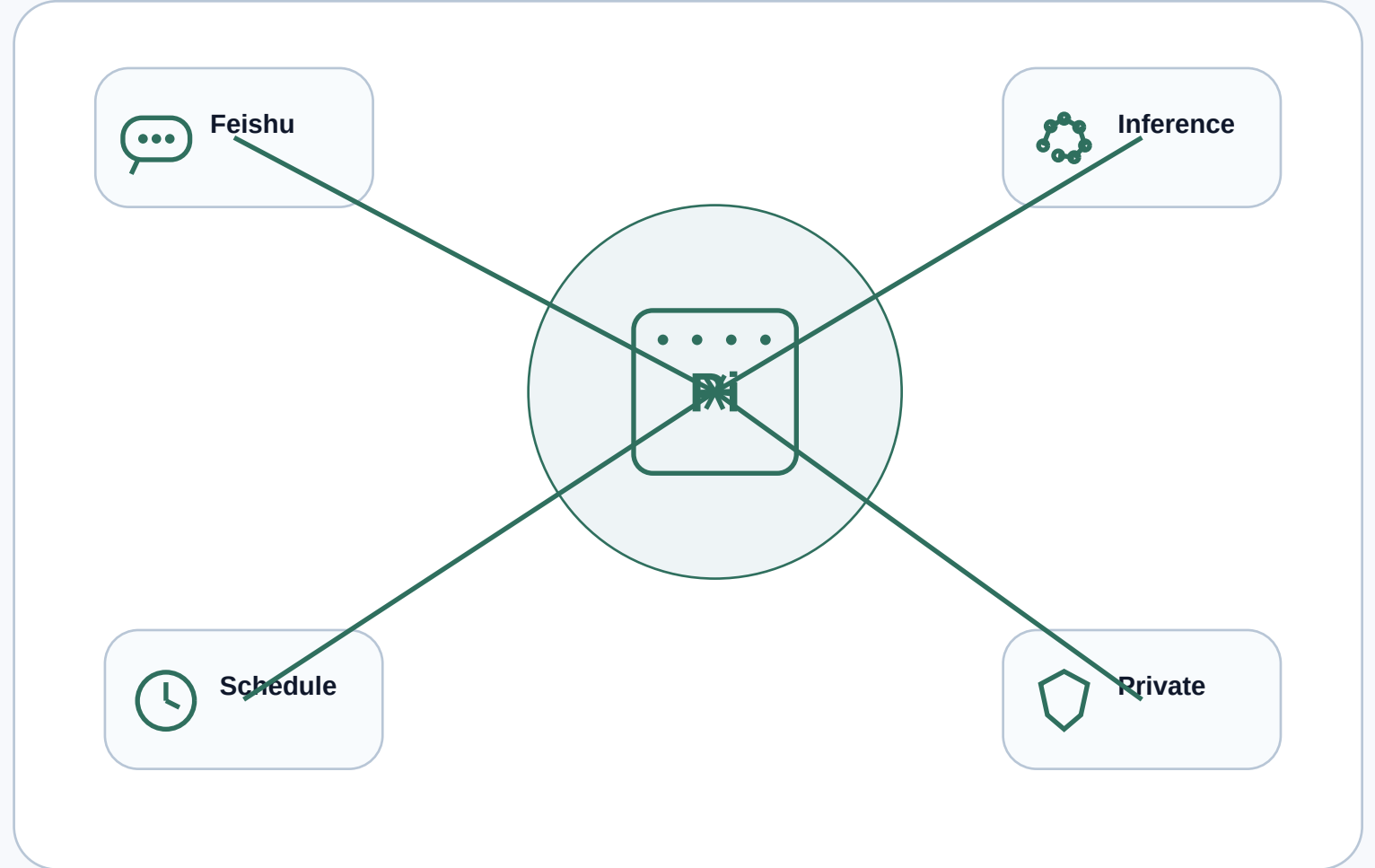


Edge Task Hub

Resource-aware Edge AI automation on Raspberry Pi.

- Feishu is only the message channel.
- Raspberry Pi is the intelligence boundary.
- Local inference, scheduling, fallback, and monitoring happen on the edge device.



What We Must Prove

This is an Edge AI course project, not a general chatbot project.

- Focus on setup, results, discussion, and analysis.
- Explain every result through edge constraints.
- Claim resource-aware local intelligence, not just LLM usage.



Setup

real Pi runtime



Results

measured tradeoffs



Analysis

edge constraints

Course target: defend the EDGE part of AI

Why Edge AI?

Cloud AI can ignore constraints that matter on a Raspberry Pi.

- Privacy: private document text should stay on the device.
- Memory and storage: model choice must fit real Pi limits.
- Latency policy: reminders must be punctual; summaries can wait.
- Reliability: the system still needs useful behavior when an LLM fails.



Privacy

edge constraint



Memory

edge constraint



Storage

edge constraint



Latency

edge constraint



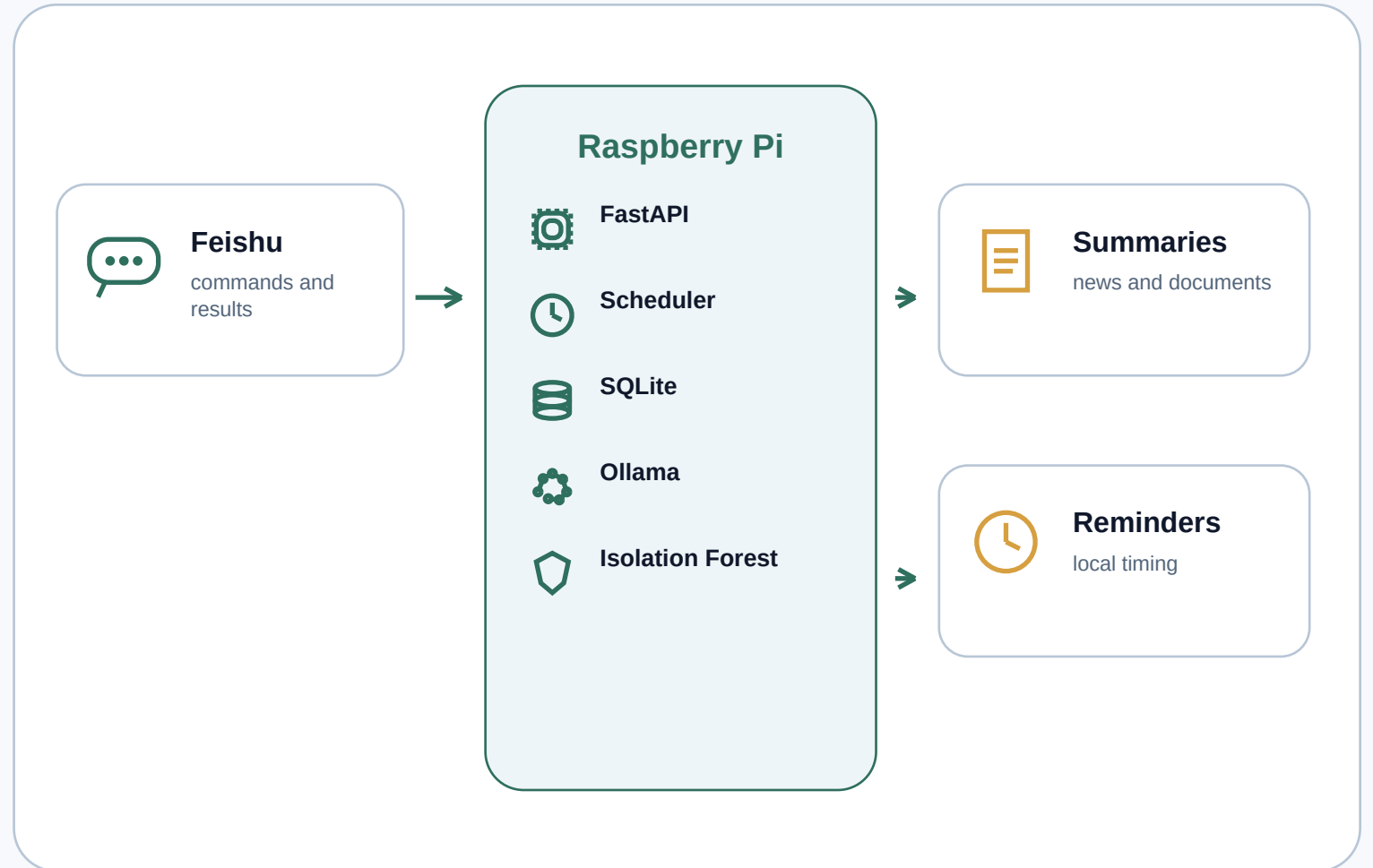
Reliability

edge constraint

System Architecture

Feishu is the interface; the Raspberry Pi owns the local decision path.

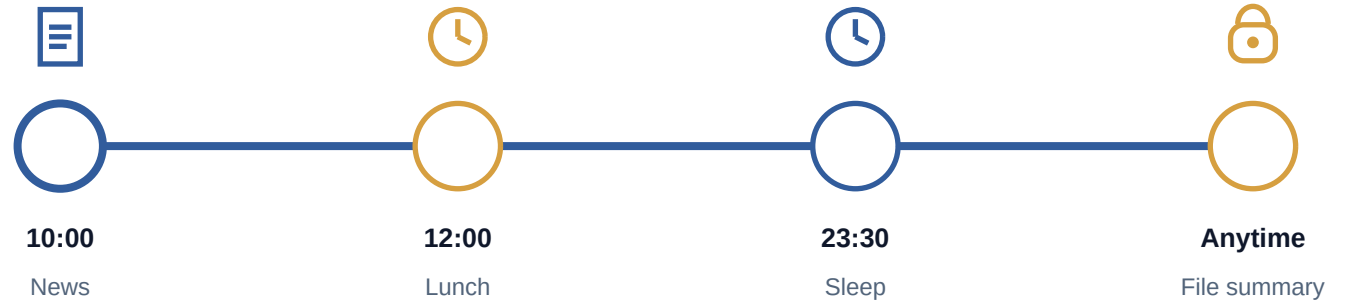
- FastAPI exposes local APIs and Feishu callbacks.
- APScheduler controls reminder and summary timing.
- SQLite records tasks, runs, events, and metrics.
- Ollama and Isolation Forest provide local intelligence.



User Workflows

The project is a useful edge system, not only a model benchmark.

- Daily previous-day news summary: 0 10 * * * local time.
- Built-in lunch and sleep reminders: 12:00 and 23:30.
- User-created reminders through Feishu text commands.
- Private document summaries from Feishu file events.



Experimental Setup

Experiments were run on the deployed Raspberry Pi, not only on a laptop.

- Runtime stack: FastAPI, APScheduler, SQLite, Ollama, and scikit-learn.
- LLM candidates: qwen3.5:2b, qwen3.5:0.8b, qwen2.5:0.5b, qwen3:1.7b, and qwen3:0.6b.
- Non-LLM intelligence: deterministic rules fallback and Isolation Forest.



Device

Raspberry Pi



Application

FastAPI / APScheduler / SQLite



AI

Ollama / rules / Isolation Forest



Messaging

Feishu webhook and callback

Model Selection Journey

Changing models was the experiment: each model tested a different edge constraint.

- 2B candidate tested quality but failed deployability.
- 0.8B quantized model tested memory reduction.
- 0.5B model tested speed but lost quality.
- Final design routes by edge job.

qwen3.5:2b

too heavy

qwen3.5:0.8b

quantized but slow

qwen2.5:0.5b

fast but weak

qwen3:1.7b + 0.6b

hybrid final

Model Results

The final model choice balances quality, speed, memory, and deployability.

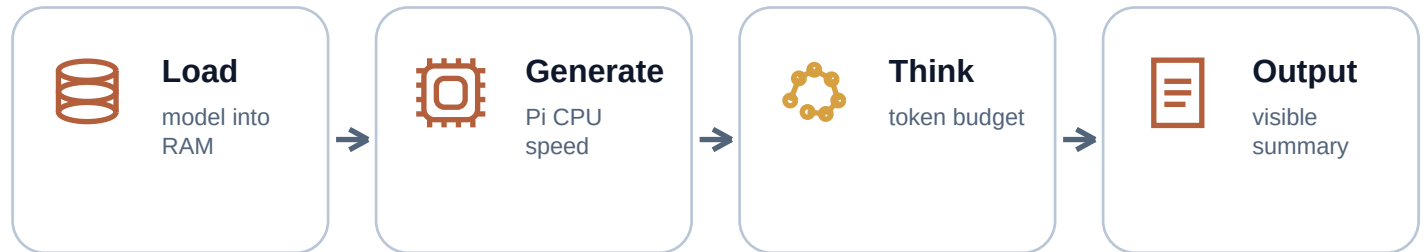
- qwen3:1.7b is selected for waitable summaries.
- qwen2.5:0.5b is fast but weaker.
- qwen3.5:0.8b is quantized but still slow.
- qwen3.5:2b is rejected before final inference.



Why qwen3.5:0.8b Was Rejected

Quantization helps model size, but it did not solve this edge workload.

- Recorded cold-start runs were about 52 to 75 seconds.
- Generation speed was about 2 tokens/s.
- Visible output was empty or length-limited.
- Likely causes: cold loading, Pi CPU speed, memory pressure, and thinking-token budget.



Observed symptom

52-75 seconds, about 2 tokens/s, empty or length-limited visible output

Why Non-LLM Models Matter

Edge AI should use the smallest reliable intelligence for each local job.

- Rules fallback is deterministic when Ollama is unavailable.
- Isolation Forest is better than an LLM for CPU, memory, disk, temperature, and network metrics.
- APScheduler plus SQLite gives local timing control for reminders.
- This is hybrid Edge AI, not only LLM Edge AI.



LLM

language summaries



Rules

deterministic fallback



Isolation Forest

device metrics



Scheduler

local timing

Privacy and Automation Results

Private processing stays local while Feishu only delivers commands and results.

- Feishu text can create scheduled reminders.
- Feishu files can trigger private document summaries.
- Document text is extracted and summarized locally.
- Verified checks cover Feishu parsing, slow-summary acceptance, document extraction, and English-only materials.



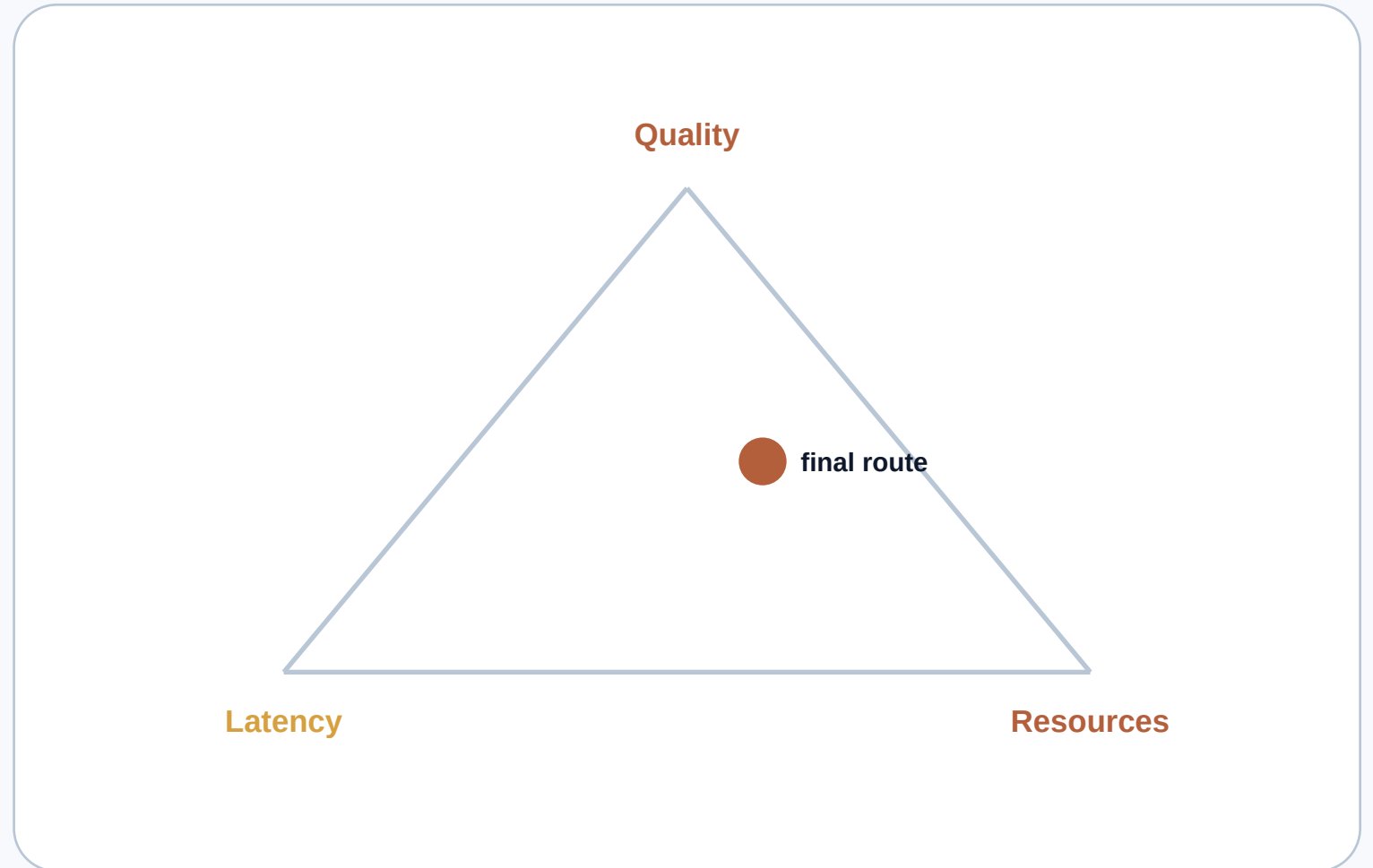
Privacy boundary

Raw document text stays on Raspberry Pi

Discussion and Analysis

The strongest project argument is the edge tradeoff, not only the generated text.

- Daily summaries can wait for qwen3:1.7b quality.
- Reminders need local timing reliability.
- Private files need local processing.
- Device health needs lightweight local ML.
- One best model does not exist for all edge tasks.



Limitations and Conclusion

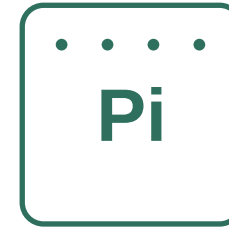
Edge Task Hub is a real Edge AI system with honest boundaries.

- Feishu public callback configuration is still required.
- Local LLM cold starts remain slow.
- Scanned PDFs need OCR in future work.
- Final claim: useful AI can run under edge constraints by choosing the right local model for each job.

Private

Reliable

Measured



Right local model for each edge job